Rational Number Project

Level 2 / Lesson 2 / Overview

Students are shown how to use cuisenaire rods to represent fractions. They relate this new model to a familiar one: fraction circles.

Materials

- Fraction Circles for teacher
- Cuisenaire rods for students and teacher
- Student Pages A-D

Teaching Actions

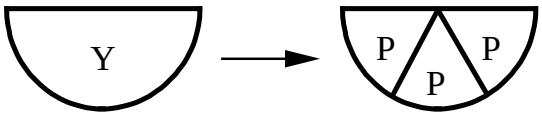
1. Start the lesson by asking students to sort through their sets of cuisenaire rods. Ask questions like:

   a) How many greens equal 1 blue?
   b) How many whites equal 1 dark green?
   c) 3 reds is the same length as what rod?
   d) If the white = 1, what is the value of an orange and yellow train?

2. Explain that today they will learn to use the rods to model fraction amounts.

3. Use the fraction circles to model the fraction 2/3 at the overhead.

   Say: to show 2/3 with the circles, I first decide on a unit. If this yellow piece is my unit, to show 2/3 I first divide it into 3 equal parts:



   2/3 is 2 of the 3 pinks. These 2 pinks are 2/3 of the yellow piece.

Comments

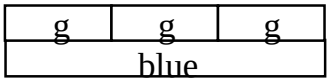
1. Students are introduced to a new model for fractions (cuisenaire rods) by relating this model to a familiar one (circles). This is a translation from one physical model to another. Seeing similarities between models helps students abstract concepts.

Teaching Actions

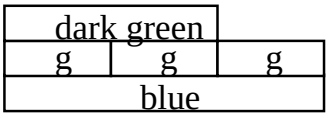
4. Now ask students to show 2/3 with their rods. Leading questions may include:

   a) What units could we use?
   b) What must we be able to do with the unit? [Divide it into 3 equal parts]
   c) Let's try a yellow rod. Can we partition it into 3 equal parts using smaller rods? [No]
   d) Let's try a blue rod. Find 3 of one color that equals the blue rod. [3 g = 1 bl]

5. Use the blue rod to model how to partition a rod into equal parts.



   Say: I want to highlight 2 of the 3 greens in some way. I could just pick them up in my hands, or with a picture I could shade in 2 of 3 equal parts. I have a special way to do this with rods. To show 2 of 3 equal parts, find one rod that covers 2 of the 3 greens. Since the dark green equals 2 greens, the dark green is 2/3 of the blue.



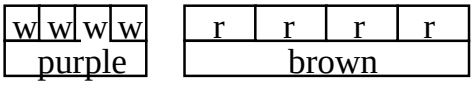
6. Let's look at the circular and cuisenaire models for the fraction 2/3. How are they alike? Different?

7. Summarize that in each case you:
   a) Decide on a unit.
   b) Divide the unit into 3 equal parts.
   c) In some way designate 2 of the 3 equal parts.

Comments

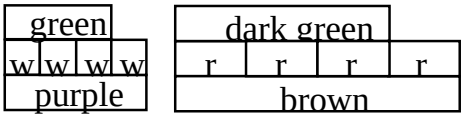
2. The following shows how to show fractions using the rods: Example: 3/4

   a) Find a unit. In this case, purple or brown can be a unit since both can be divided into 4 parts using a smaller rod.

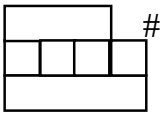


   b) Divide the unit into equal parts (see above).

   c) To show 3 of 4 equal parts find one rod that equals 3 whites or 3 reds and place on top of model.



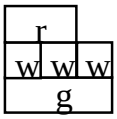
3. Notice that there are 3 parts to each model:

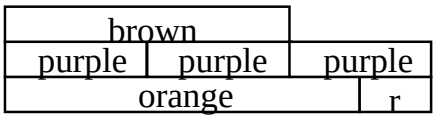


# of equal parts considered
partitions
unit

Teaching Actions

8. Ask if you could show 2/3 with circles using a different unit. Have students find a different unit with the cuisenaire rods to show 2/3.





9. Model other fractions using rods. Ask students to explain how their models show each fraction. Initially suggest the unit and in later examples have students generate the unit.

$\frac{3}{4}$ brown

$\frac{1}{3}$ orange & red train

$\frac{1}{2}$ blue* see Comment 5

$\frac{2}{5}$ student generates unit

$\frac{4}{6}$ student generates unit

$\frac{1}{4}$ student generates unit

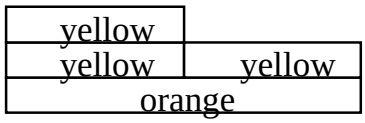
10. End the initial discussion by asking this question: Can you use the black rod as the unit for 1/3? Does that mean the black rod can't be divided into thirds?

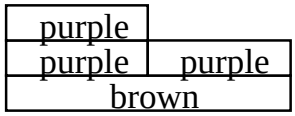
11. Student Pages A-D provide practice. Select the amount you feel is needed for students.

Comments

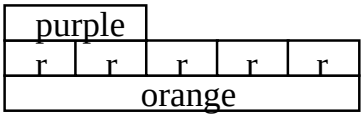
4. Other examples:

$\frac{1}{2}$

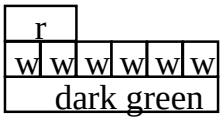


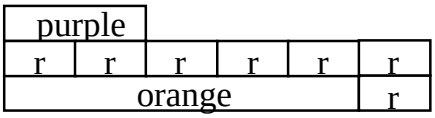


$\frac{2}{5}$



$\frac{2}{6}$





5. The blue rod represents an odd number. The solution to this problem exists, but it cannot be demonstrated with cuisenaire rods.

6. There are limits to physical models. To divide the black rod into thirds you would need a saw; you can't show it with the rods. Each third would be greater than 1 red, but less than 1 green.

Name _____

Lesson 2
Student Page A

Directions:

Answer the following questions for each figure:

a) Word name for the fraction shown.

b) Symbol name for the fraction shown.

c) _____ of _____ equal-sized parts are covered.

d) Write a sentence using the middle layer to describe a relationhsip between the top layer and bottom layer.

The first problem is worked for you.

1.

brown				
r	r	r	r	r
orange				

a) 4-fifths

b) $\frac{4}{5}$

c) 4 of 5 equal-sized parts are covered.

d) 1 brown is $\frac{4}{5}$ of an orange.

2.

purple			
r	r	r	r
brown			

a)

b)

c)

d)

3.

purple		
r	r	r
dark green		

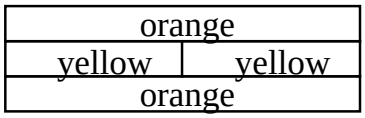
a)

b)

c)

d)

4.



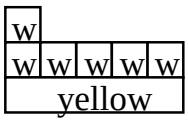
a)

b)

c)

d)

5.



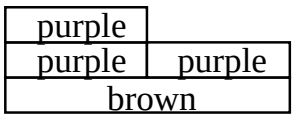
a)

b)

c)

d)

6.



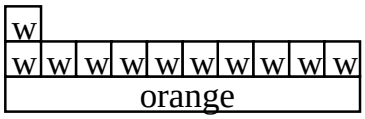
a)

b)

c)

d)

7.



a)

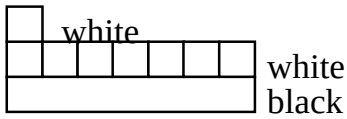
b)

c)

d)

Directions:
If you can, use the rod color given as a unit. If you can't show that fraction using the given rod as the unit, write "impossible". Draw a picture of your rod display. Write the color of the rods you use. The first one is done for you.

1. Unit: 1 black

show $\frac{1}{7}$ 

show $\frac{1}{3}$ Impossible with black as a unit

2. Unit: 1 blue

show $\frac{1}{9}$

show $\frac{1}{3}$

show $\frac{1}{2}$

3. Unit: 1 dark green

show $\frac{1}{6}$

show $\frac{1}{2}$

(Continued)

Unit: 1 dark green

show $\frac{1}{3}$

show $\frac{1}{4}$

4. Unit: 1 orange

show $\frac{3}{5}$

show $\frac{6}{10}$

show $\frac{2}{3}$

5. Unit: 1 blue

show $\frac{2}{3}$

show $\frac{3}{4}$

show $\frac{4}{9}$